Research Statement

My goal is to develop more trustworthy and equitable machine learning systems in social applications such as healthcare and privacy. Over the past three years at Massachusetts General Hospital (MGH), I have gained experience in clinical machine learning (ML) in medical imaging through both fundamental and translational research. Previously, I worked on applying ML to a diverse range of applications such as supercomputing, satellite imagery, and computational biology during my research experiences at the University of Georgia (UGA), NASA, and Oak Ridge National Laboratory (ORNL).

1 Experience

While an undergraduate at UGA, I contributed to projects in the areas of computational biology, political science, and geographic information systems (GIS). Under supervision of Dr. Shannon Quinn at UGA and with collaborators at the Children's Hospital at the University of Pittsburgh Medical Center, I developed a fully automatic pipeline to detecting abnormal ciliary motion in microscopy video. The methods I developed allowed for a faster and more standardized approach to detect abnormal motion of cilia to better study ciliopathy, a rare genetic disorder associated with birth effects. I published this work in two conference papers [3, 8].

To further develop my research skills, I worked as a research assistant for Dr. Jason Anastasopoulos at UGA and Dr. Teppei Yamamoto at MIT to build a web scrapper, which cycled IP addresses to collect video metadata, to study the influence of recommender algorithms on political polarization in a popular video streaming platform. I also participated in the NASA DEVELOP National Program at UGA under the supervision of Dr. Marguerite Madden at the Center for Geospatial Research, where I analyzed NASA satellite imagery to study vegetation damage caused by Hurricane Irma. To forecast water quality in the surrounding area of Biscayne Bay, I developed a time-series model to predict over water quality attributes, such as chlorophyll and turbidity, from 10 years of ground station measurements. Our research efforts helped to inform environmental infrastructure policy for our project partners in the Public Works Department of the City of Miami Beach [1].

I then interned at the U.S. Department of Energy's ORNL in Dr. Arvind Ramanathan's computational chemistry group, where I investigated parallel programming algorithms to scale neural networks for molecular dynamics simulations on the Summit supercomputer [2]. During this internship, I taught myself tensor algebra and CUDA programming to be able to contribute to an internal deep learning framework to enable exascale deep learning for basic science applications.

After graduating I joined the Center for Clinical Data Science (CCDS) of MGH & Brigham and Women's Hospital (BWH) as a data scientist, where I led technical development for several commercial projects such as the detection of stroke, measurement of aortic aneurysms, grading of lumbar stenosis, localization of cervical fractures, and the detection of kidney stones. From

my experiences in clinical machine learning, I wrote papers on the challenges of developing and deploying medical AI in the real world to help bridge the gap between research and translational application [4, 5].

I also joined the Quantitative Translational Imaging in Medicine laboratory under Dr. Jayashree C. Kalpathy at the Athinoula A. Martinos Center for Biomedical Imaging and MGH, where I led a project leveraging federated learning for retinopathy of prematurity, the leading cause of blindness in infants. My work focused on building a collaborative learning framework that allows training across medical institutions without the need to share patient data while better maintaining patient privacy. Among my results, I showed that federated learning performs as well as centrally hosted patient data and also that such approaches can be used to monitor local disease epidemiology and clinician diagnostic paradigms. Our work develops one of the first instance of federated learning in ophthalmology and demonstrates the performance advantage of collaborating across multiple hospitals to aid diagnosis of low-prevalence diseases.

2 Research Interests

One perennial challenge in machine learning is the trustworthy deployment into real-world environments with human users. My previous work, which focused on developing medical imaging machine learning systems for clinical deployment, motivated me to research how uncertainty can be incorporated into current deep learning methods to improve fairness and transparency in critical medical decision-making tools. Using a large national mammography dataset for breast cancer screening, I analyzed patient race and scanner type subgroups and found that deep learning classifiers have disparate performance on certain minority groups. These results motivated me to further investigate how epistemic uncertainty measures could be used to calibrate users to varying levels of confidence on different subgroups and resulted in a paper at the International Conference on Machine Learning (ICML) workshop on Interpretable Machine Learning in Healthcare [6]. In follow-up work, I conducted field surveys with clinicians to inform the design of fairer clinical machine learning workflows with robust confidence guarantees using distribution-free uncertainty methods, specifically conformal prediction sets, that could facilitate better human-machine interaction. This work on fair conformal predictors was accepted to the AI for Social Impact track at the Association for the Advancement of Artificial Intelligence (AAAI) conference [7].

In my previous research experiences, I have had the opportunity to solve a variety of computational problems – from analyzing hurricanes in satellite imagery to studying the microscopic motion of cilia to creating AI software medical devices for stroke. As I worked closely with domain experts in fields such as political science, climate science, and radiology, I have become interested in the social impact of computational tools and algorithmic systems. For example, in medicine, many FDA cleared AI devices are black-boxes to clinicians, which hamper transparency (biases may exist in the model) and lower user confidence (without robust uncertainty estimates). Issues such as privacy and fairness in machine learning have the potential to cause societal harm and reinforce disparities between protected demographic groups. I believe my various experiences solving applied machine learning problems in a diverse range of applications will allow me to develop new techniques and frameworks to better impute human-aligned values into future machine learning systems.

References

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